

1.

$$\begin{aligned} f + g &= t^3 + t^2 \\ f * g &= t^5 + t^4 + 1 \\ f^2 &= t^6 + t^2 + 1 \end{aligned}$$

2. From definition of unit:

$$uv = 1$$

then

$$\begin{aligned} \deg(uv) &= 0 \\ \deg(uv) &= \deg(u) + \deg(v), \deg(u) \geq 0, \deg(v) \geq 0 \implies \deg(u) = 0, \deg(v) = 0 \end{aligned}$$

Therefore units of $\mathbb{F}[t]$ are polynoms of form $f(t) = c$ where c is a unit of $\mathbb{F}$, since $\mathbb{F}$ is a field and $c \neq 0$ and by definition of a field c has an inverse. $F[t]^\times = F^\times = F \setminus 0$

3.

$$\begin{aligned} t^2 &= tt \\ t^2 + 1 &= (t - 1)(t + 1) \\ t^2 + t &= t(t + 1) \\ f(t) &= t^2 + t + 1, f(0) = 1, f(1) = 1 \text{ --- irreducible} \end{aligned}$$

4.

$$\begin{aligned} f \mid g &\implies \exists h: g = fh \\ \deg(g) &= \deg(f) + \deg(h), \deg(f) \geq 0, \deg(h) \geq 0 \implies \deg(g) \geq \deg(f) \end{aligned}$$

If $\deg(f) = \deg(g) = d$ then $\deg(h) = 0, h(t) = c$ and:

$$\begin{aligned} f(t) &= a_0 + a_1t + \dots + a_dt^d \\ g(t) &= b_0 + b_1t + \dots + b_dt^d \\ \forall i: b_i &= ca_i \\ g \neq 0, f \neq 0 &\implies c \neq 0 \text{ and } f, g \text{ --- associates} \end{aligned}$$

5. Assume $f(t) = t$ is reducible, then $f = gh, \deg(g) < \deg(f), \deg(h) < \deg(f)$. Then since $\deg(f) = 1$, then $\deg(g) = \deg(h) = 0, h \in \{0, 1\}, g \in \{0, 1\}$. $0*0 = 0 \neq t, 0*1 = 0 \neq t, 1*0 = 0 \neq t, 1*1 = 1 \neq t$. Exactly same logic applies to $f(t) = t + 1$.

Therefore since $t \mid t^2 + t, (t + 1) \mid t^2 + t$ then t and $(t + 1)$ are factors of $t^2 + t$.

Since

$$\deg f \in \{2, 3\} \implies f \text{ reducible over } F \iff f \text{ has a root in } F.$$

then because for $\mathbb{F}_2 = \{0, 1\}$ and $f(t) = t^2 + t + 1$

$$\begin{aligned} f(0) &= 1 \\ f(1) &= 1 \end{aligned}$$

therefore $f(t)$ doesn't have a root, it's not reducible.